

Lecture 9: Matrix Algebra

Mathematics 2 (INF 1004)

2026

1 Introduction to Matrices

A matrix A is a two-dimensional rectangular array of elements arranged in rows and columns.

- A matrix with m rows and n columns is an $m \times n$ matrix.
- An element in the i^{th} row and j^{th} column is denoted as a_{ij} .

1.1 Special Types of Matrices

- **Square Matrix:** A matrix where $m = n$.
- **Identity Matrix (I):** A square matrix with 1s on the main diagonal and 0s elsewhere.
- **Zero Matrix:** A matrix where all elements are zero.

2 Basic Matrix Operations

- **Addition/Subtraction:** Performed element-wise; matrices must be the same size.
- **Scalar Multiplication:** Multiply every element by the scalar λ .
- **Transpose (A^T):** Formed by interchanging rows and columns.
- **Matrix Multiplication:** Possible only if the columns of A equal the rows of B .

2.1 Transpose (A^T)

The transpose is formed by reversing the rows and columns of a matrix.

- **Symmetric Matrix:** A square matrix where $A^T = A$.
- **Skew-Symmetric Matrix:** A square matrix where $A^T = -A$.

2.2 Matrix Multiplication

Two matrices A and B can be multiplied (as AB) only if the number of columns in A equals the number of rows in B .

- If A is $m \times p$ and B is $p \times n$, the resulting matrix AB is $m \times n$.
- Multiplication is non-commutative: $AB \neq BA$, but associative and distributive.

3 Determinants, Minors, and Cofactors

For a square matrix A :

- **Minor (M_{ij}):** The determinant of the sub-matrix formed by deleting the i^{th} row and j^{th} column.
- **Cofactor (C_{ij}):** The signed minor, calculated as $C_{ij} = (-1)^{i+j}M_{ij}$.

4 Numerical Examples

4.1 Matrix Multiplication

Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 3 \end{bmatrix}$.

$$\bullet AB = \begin{bmatrix} (1 \cdot 2 + 1 \cdot 0 + 0 \cdot 1) & (1 \cdot 0 + 1 \cdot 1 + 0 \cdot 3) \\ (2 \cdot 2 + 0 \cdot 0 + 1 \cdot 1) & (2 \cdot 0 + 0 \cdot 1 + 1 \cdot 3) \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$$

4.2 3x3 Determinant Expansion

For $A = \begin{bmatrix} 3 & 4 & -1 \\ 2 & 0 & 7 \\ 1 & -3 & -2 \end{bmatrix}$, expanding along Row 1:

$$\begin{aligned} |A| &= 3 \begin{vmatrix} 0 & 7 \\ -3 & -2 \end{vmatrix} - 4 \begin{vmatrix} 2 & 7 \\ 1 & -2 \end{vmatrix} + (-1) \begin{vmatrix} 2 & 0 \\ 1 & -3 \end{vmatrix} \\ &= 3(0 - (-21)) - 4(-4 - 7) - 1(-6 - 0) \\ &= 3(21) - 4(-11) + 6 = 63 + 44 + 6 = 113 \end{aligned}$$

4.3 Adjoint and Inverse Calculation

To find A^{-1} , we first find the matrix of cofactors, then the adjoint.

Step 1: Calculate Cofactors (C_{ij})

$$\begin{aligned} C_{11} &= + \begin{vmatrix} 0 & 7 \\ -3 & -2 \end{vmatrix} = 21 & C_{12} &= - \begin{vmatrix} 2 & 7 \\ 1 & -2 \end{vmatrix} = 11 & C_{13} &= + \begin{vmatrix} 2 & 0 \\ 1 & -3 \end{vmatrix} = -6 \\ C_{21} &= - \begin{vmatrix} 4 & -1 \\ -3 & -2 \end{vmatrix} = 11 & C_{22} &= + \begin{vmatrix} 3 & -1 \\ 1 & -2 \end{vmatrix} = -5 & C_{23} &= - \begin{vmatrix} 3 & 4 \\ 1 & -3 \end{vmatrix} = 13 \\ C_{31} &= + \begin{vmatrix} 4 & -1 \\ 0 & 7 \end{vmatrix} = 28 & C_{32} &= - \begin{vmatrix} 3 & -1 \\ 2 & 7 \end{vmatrix} = -23 & C_{33} &= + \begin{vmatrix} 3 & 4 \\ 2 & 0 \end{vmatrix} = -8 \end{aligned}$$

Step 2: Form the Adjoint Matrix ($adj(A)$) The adjoint is the transpose of the cofactor matrix:

$$adj(A) = \begin{bmatrix} 21 & 11 & -6 \\ 11 & -5 & 13 \\ 28 & -23 & -8 \end{bmatrix}^T = \begin{bmatrix} 21 & 11 & 28 \\ 11 & -5 & -23 \\ -6 & 13 & -8 \end{bmatrix}$$

Step 3: Calculate the Inverse (A^{-1}) Using $|A| = 113$:

$$A^{-1} = \frac{1}{113} \begin{bmatrix} 21 & 11 & 28 \\ 11 & -5 & -23 \\ -6 & 13 & -8 \end{bmatrix}$$

5 Systems of Linear Equations (SLE)

A system can be represented as $AX = B$. If $|A| \neq 0$, the solution is $X = A^{-1}B$.